



Fig. 3: ATmega168 Development Stick

used to display the calculated result. Port pins PB3 through PB0 of IC1 are connected to data pins D4 through D7, and port pins PB4 and PB5 are used to provide control signals EN and RS to LCD1, respectively. R/W pin is grounded permanently, as shown in Fig. 2. Switch S1 is used to manually reset microcontroller IC1.

Postfix notation. The postfix notation method was introduced in 1954 and was first used in desktop calculators by HP in 1963. However, now almost all the calculators deploy it to perform calculation of user inputs. In postfix notation the operators follow their operands; for instance, to add 4 and 6, one would write $4\ 6\ +$ rather than $4 + 6$. If there are multiple operations, the operator is given im-

TABLE II
Step-by-Step Evaluation of Regular Expression $3+2\times 4/(1-5)$

Token	Action	Output string	Stack	Remarks
3	Add to output	3		
+	Push to stack	3	+	
2	Add to output	3 2	+	
$\times$	Push to stack	3 2	* +	* has higher precedence than +
4	Add to output	3 2 4	* +	
/	Pop stack to output	3 2 4 *	+ /	/ and * have same precedence
	Push to stack	3 2 4 *	/ +	
(	Push to stack	3 2 4 *	(/ +	
1	Add to output	3 2 4 * 1	(/ +	
-	Push to stack	3 2 4 * 1	- (/ +	
5	Add to output	3 2 4 * 1 5	- (/ +	
)	Pop stack to output	3 2 4 * 1 5 -	(/ +	Repeated until next (is found
	Pop stack	3 2 4 * 1 5 -	/ +	Discard closing parenthesis
end	Pop stack to output	3 2 4 * 1 5 - /	+	
	Pop stack to output	3 2 4 * 1 5 - / +		

PARTS LIST

Semiconductors:

- IC1 - ATmega168 microcontroller
- IC2 - FT232RL serial-to-UART converter

- LED1-LED3 - 5mm LED
- LCD1 - 16 $\times$ 2 LCD
- T1 - BC548 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3, R4 - 680-ohm
- R2 - 1-kilo-ohm
- R5 - 100-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 0.1 μ F ceramic
- C2-C4 - 0.01 μ F ceramic

Miscellaneous:

- CON1 - USB B-type connector
- S1 - Tactile switch

mediately after its second operand. So the expression $1 - 4 + 6$ written in conventional notation would be written as $1\ 4 - 6\ +$ in postfix. That means, first subtract 4 from 1 and then add 6 to that.

An advantage of postfix is that it obviates the need for parentheses, which are required by normal notations, and thus removes ambiguity from the formulae. For instance, $1 - 4 \times 6$ can also be written $1 - (4 \times 6)$ and it is quite different from $(1 - 4) \times 6$. In postfix, the former could be written as $1\ 4\ 6\ \times\ -$, which unambiguously means $1\ (4\ 6\ \times)\ -$, whereas the latter could be written $1\ 4 - 6\ \times$ or $6\ 1\ 4 - \times$. As the notation result is always

context-free, once an equation is converted to postfix notation, it becomes easier for a computer to evaluate the same using outside-in evaluation sequence.

Working of the circuit is simple. Once the program is running, the input equation to be solved can be provided using hyper-terminal or other similar software like X-CTU. The equation and the calculated result will be displayed on the LCD. (Refer construction and testing section for details.)

Software

The program code is written in 'C' language and compiled using AVR Studio 6.0. The hex file is burnt into the microcontroller using a suitable programmer.

Postfix algorithm. The method is stack-and-queue dependant and uses a stack for storing functions (*aka* operators) and a simple queue to hold numbers (*aka* operands). Due to the nature of operations performed in this algorithm, it is also called a 'shunting yard algorithm' since the operation resembles a railroad shunting-yard methodology. Table II shows step-by-step evaluation of a regular expression using postfix algorithm.

Once parsing is done, rest of the part is much easier to perform. For